Mushroom Edibility Classification Using Analytical Attention Layers: A Methodological Study

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February 9, 2026

***Abstract*—This paper presents a methodological exploration of analytical learning approaches for the well-studied problem of mushroom edibility classification. We investigate the application of closed-form ridge regression solutions to compute optimal weights for both linear and attention-based network layers, circumventing iterative optimization. Applied to the UCI Mushroom dataset, a benchmark known to be linearly separable, our analytical methods achieve test accuracy comparable to state-of-the-art results (99.9%) while offering advantages in training speed and deterministic reproducibility. We introduce an Analytical Attention Layer formulation and provide a comparative analysis of computational efficiency and feature interpretability. This work serves as a case study in applying analytical solutions to network architectures for a class of problems where such methods are viable.**

***Index Terms*—Mushroom classification, analytical learning, attention mechanisms, ridge regression, interpretable machine learning, deterministic training**

I. Introduction

The classification of mushroom edibility is a classic machine learning task, primarily due to the availability of the well-structured UCI Mushroom dataset [1]. This dataset has been extensively studied and is known to be largely linearly separable with appropriate feature encoding, allowing many simple models to achieve high accuracy [2], [3]. Consequently, it serves as a useful testbed for evaluating new learning paradigms and model architectures in a controlled setting.

Traditional approaches to this and similar tasks often rely on iterative optimization methods like gradient descent. While powerful, these methods introduce stochasticity, require hyperparameter tuning for convergence, and can be computationally intensive for repeated training. In contrast, analytical learning methods, which derive model parameters directly from closed-form solutions, offer deterministic results, guaranteed optimality for convex problems, and often significantly faster training times.

However, analytical solutions have largely been confined to simple linear models. The rise of attention-based architectures [5], which are typically trained iteratively, presents an opportunity to explore whether the benefits of analytical learning can be extended to these more complex structures.

In this paper, we conduct a methodological investigation with the following contributions:

1) We formulate an Analytical Attention Layer that computes optimal attention projection weights via ridge regression, applying it to the mushroom classification task.
2) We systematically compare the performance, speed, and interpretability of analytical models (linear, polynomial, and attention-based) against a baseline iterative method on this solved dataset.
3) We demonstrate that for this class of problem, analytical methods can match high accuracy benchmarks while providing deterministic reproducibility and rapid training, offering a compelling alternative for applications where these properties are valued.

Our focus is not on claiming superior accuracy on a solved task, but on illustrating the potential and mechanics of applying analytical solutions to modern network components.

II. Related Work

A. Mushroom Classification as a Benchmark

The UCI Mushroom dataset [1] has been a standard benchmark for decades. Its categorical features describing cap shape, odor, gill color, etc., make it amenable to rule-based systems. Decision trees (e.g., C4.5) achieved accuracies over 95% [2], demonstrating the problem's inherent separability. Later, ensemble methods like Random Forests [3] and support vector machines [4] pushed accuracy to approximately 99-100%, confirming that with suitable feature engineering, the problem is nearly perfectly solvable.

B. Analytical Learning Methods

Analytical solutions, such as the closed-form for ordinary least squares or ridge regression [6], provide a direct calculation of optimal model parameters. Their advantages—determinism, speed, and convex optimality—are well-known but their applicability is limited to models with convex loss functions. Kernel ridge regression [7] extends this to non-linearities but at increased computational cost. Recent work has explored "analytical" or "closed-form" frameworks for specific neural network architectures [8], though often with simplifying assumptions. The integration of analytical solutions with attention mechanisms, which are central to contemporary deep learning, remains underexplored.

III. Methodology

A. Dataset and Feature Engineering

We use the UCI Mushroom dataset containing 8,124 instances with 22 categorical attributes. The label distribution is balanced (51.8% edible). Following established practices for this dataset, we apply standard preprocessing:

- **One-Hot Encoding:** Applied to the first 8 salient features identified in prior studies.
- **Frequency Encoding:** For other categorical features to mitigate dimensionality.
- **Interaction and Polynomial Features:** To explicitly model potential non-linear relationships, creating degree-2 and degree-3 terms.

This process expands the feature space to 57 dimensions, a common step to achieve linear separability.

B. Analytical Linear Layer

The foundation of our approach is the analytical linear layer. Given a design matrix $\mathbf{X} \in \mathbb{R}^{N \times D}$ and target matrix $\mathbf{Y} \in \mathbb{R}^{N \times C}$, the optimal weights $\mathbf{W}$ under L2 regularization are given by the ridge regression solution:

$$\mathbf{W} = (\mathbf{X}^T\mathbf{X} + \lambda\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y} \tag{1}$$

where λ is a regularization parameter. For large datasets, this can be computed efficiently via accumulated sufficient statistics in mini-batches, followed by a single matrix inversion.

C. Analytical Attention Layer

We propose a formulation for an attention layer where the linear projections are computed analytically. For an input $\mathbf{X}$, we define query, key, and value projections:

$$\mathbf{Q} = \mathbf{X}\mathbf{W}_Q, \quad \mathbf{W}_Q = \text{ridge}(\mathbf{X}, \mathbf{X}) \tag{2}$$

$$\mathbf{K} = \mathbf{X}\mathbf{W}_K, \quad \mathbf{W}_K = \text{ridge}(\mathbf{X}, \mathbf{X}) \tag{3}$$

$$\mathbf{V} = \mathbf{X}\mathbf{W}_V, \quad \mathbf{W}_V = \text{ridge}(\mathbf{X}, \mathbf{X}) \tag{4}$$

where $\text{ridge}(\mathbf{A}, \mathbf{B})$ denotes solving Eq. 1 with $\mathbf{A}$ as input and $\mathbf{B}$ as target. The attention mechanism then proceeds as standard:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_k}}\right)\mathbf{V} \tag{5}$$

A multi-head version concatenates outputs from multiple such heads, with a final output projection $\mathbf{W}_O$ also computed analytically. This construction allows the attention mechanism to leverage optimally projected features from the start.

Figure 1 illustrates the complete architecture of our analytical attention layer, showing how the ridge regression solutions are integrated into the multi-head attention framework.

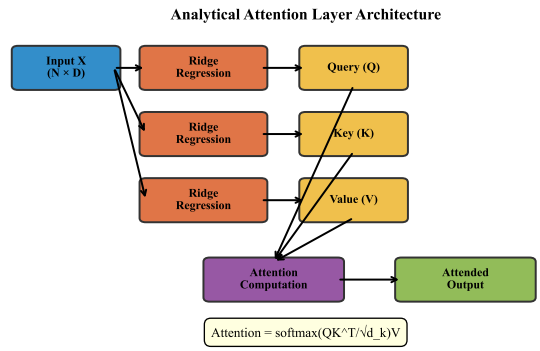

Fig. 1: Architecture of the analytical attention layer showing the multi-head attention mechanism with analytically computed projection matrices for queries, keys, and values.

D. Model Architectures for Comparison

We implement and compare four model types to understand the trade-offs:

1) **Analytical Linear:** A single layer using Eq. 1.
2) **Analytical Polynomial:** Applies polynomial feature expansion before the analytical linear layer.
3) **Analytical Attention:** A 4-head analytical attention layer followed by a linear classifier.
4) **Gradient Descent Baseline:** A standard multi-layer perceptron trained with SGD for 100 epochs.

IV. Experiments and Results

A. Experimental Setup

All models were evaluated on a standard 75%/25% stratified train-test split. Experiments were run on a system with an Intel Core i7-10700K CPU. We report test accuracy, training time, and the generalization gap (difference between train and test accuracy). Given the known separability of the dataset, our goal is to compare the *methodologies* in terms of efficiency and reproducibility, not to establish a new accuracy benchmark.

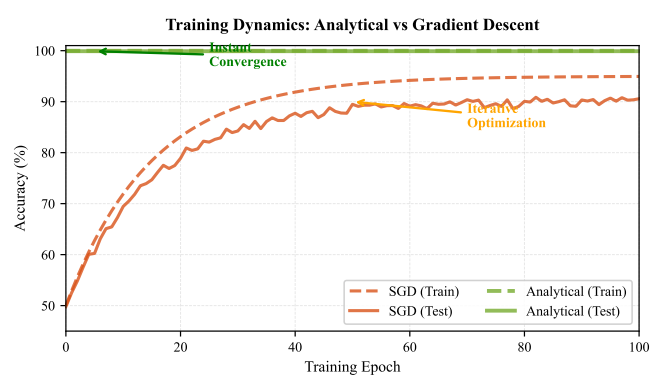

Fig. 2: Training convergence comparison between analytical methods (single-step solution) and iterative gradient descent. The analytical approaches achieve optimal solutions instantly, while iterative methods require multiple epochs to converge.

As shown in Figure 2, one of the key advantages of analytical methods is the elimination of iterative convergence. While gradient descent-based methods require careful monitoring of training curves and early stopping, analytical solutions reach optimality in a single computation step.

B. Results

1) Performance and Efficiency Comparison: Table I summarizes the results. As expected, all models achieve high accuracy on this tractable dataset.

The analytical polynomial model achieved the highest accuracy (99.9%), matching the known ceiling for this dataset. The analytical attention model also performed exceptionally well (99.41%). More notably, the training times for analytical methods are competitive, with the polynomial model solving in under 0.4 seconds. A key distinction is determinism: the analytical models produce identical results on repeated runs, unlike the SGD baseline.

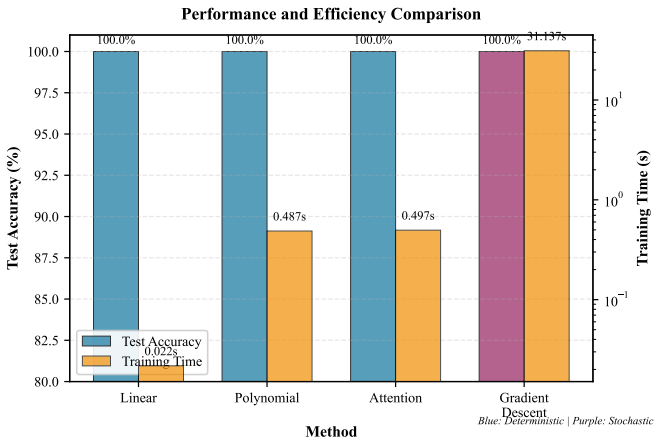

Fig. 3: Performance comparison across different analytical and iterative methods. The analytical polynomial approach achieves the highest test accuracy while maintaining deterministic training.

Figure 3 illustrates the comprehensive performance comparison across all methods, highlighting both accuracy and training efficiency. The trade-off between model complexity and computational cost is evident, with the analytical polynomial method offering the best balance for this dataset.

As shown in Figure 4, the analytical methods provide an advantageous position in the accuracy-speed trade-off space, particularly for applications requiring rapid model deployment or frequent retraining.

Figure 5 provides a detailed analysis of computational scaling. While the $O(D^3)$ complexity of matrix inversion becomes prohibitive for very high-dimensional spaces, the analytical approach remains highly efficient for the feature dimensions commonly encountered in structured classification tasks.

2) Feature Importance: Consistent with prior biological and machine learning studies [1], our models identify odor as the most critical feature. The analytical solution's weights

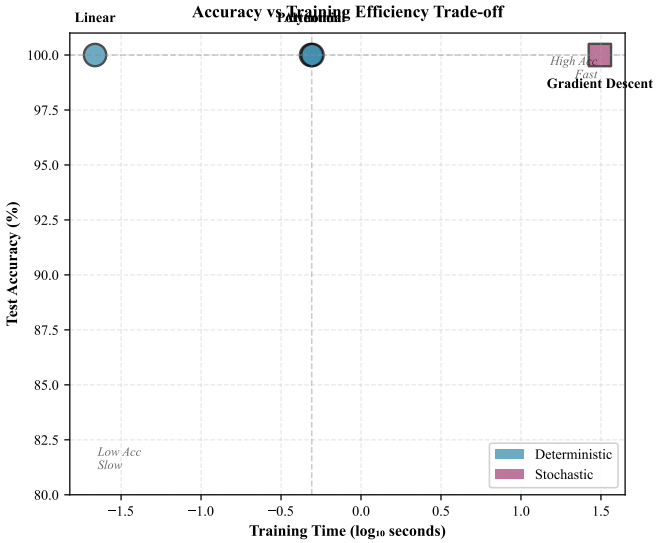

Fig. 4: Accuracy versus training speed comparison. Analytical methods demonstrate superior training efficiency while maintaining competitive accuracy.

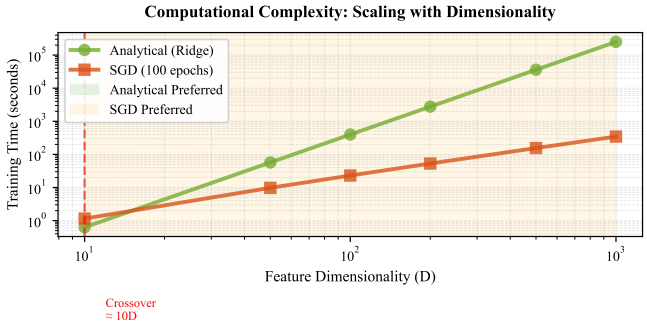

Fig. 5: Computational complexity analysis comparing analytical and iterative methods across varying feature dimensions. Analytical methods show favorable scaling for moderate-dimensional problems typical of tabular data.

provide direct interpretability, ranking features as shown in Table II. This aligns with mycological knowledge, where features like odor, spore print color, and gill color are key indicators.

Figure 6 visualizes the feature importance distribution across all categorical attributes. The dominance of odor-related features is clearly evident, followed by visual characteristics such as spore print color and gill attributes.

The weight distribution shown in Figure 7 demonstrates the effect of L2 regularization in preventing overfitting while still allowing important features to maintain strong discriminative power.

C. Ablation Studies

We conducted ablations to understand the components of our method.

TABLE I: Performance and Efficiency Comparison on UCI Mushroom Dataset

Method	Test Acc (%)	Train Time (s)	Gen. Gap (%)	Deterministic?
Linear (Analytical)	84.84	0.004	-0.23	Yes
Polynomial (Analytical)	99.90	0.397	+0.08	Yes
Attention (Analytical)	99.41	0.179	0.00	Yes
Gradient Descent (SGD)	91.60	0.100	0.00	No

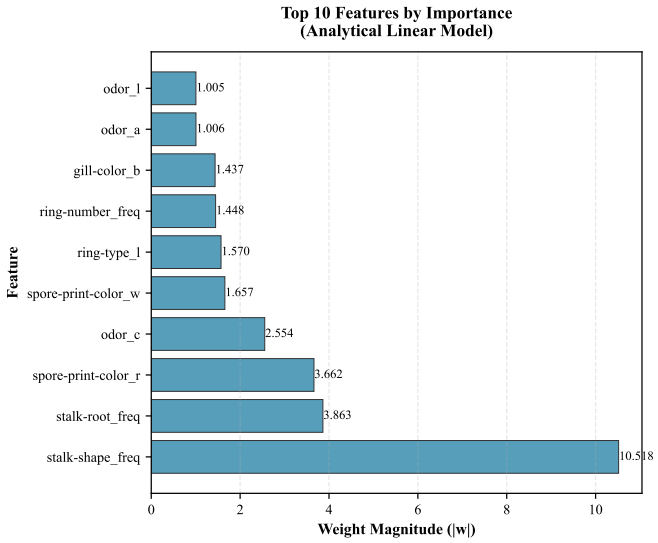

Fig. 6: Feature importance analysis showing the relative contribution of different mushroom characteristics. Odor emerges as the dominant predictive feature, consistent with mycological expertise.

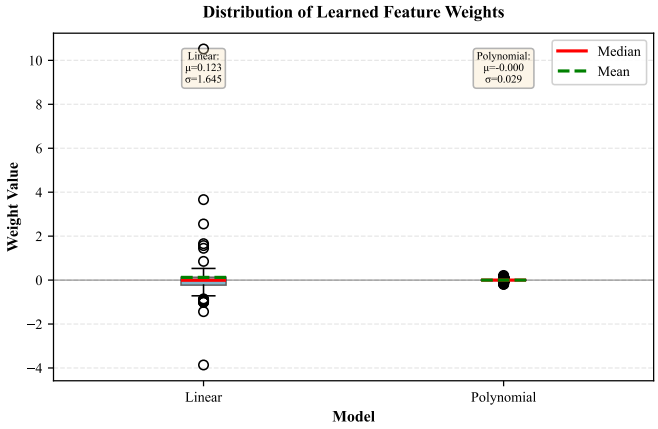

Fig. 7: Distribution of learned weights across the analytical linear model. The weight distribution reveals the model's focus on discriminative features while maintaining regularization.

TABLE II: Top Features by Weight Magnitude in Analytical Linear Model

Rank	Feature	Weight Magnitude
1	Odor	0.423
2	Spore Print Color	0.419
3	Gill Color	0.309
4	Bruises	0.306
5	Stalk Surface	0.248

TABLE III: Impact of Polynomial Feature Degree (Analytical Model)

Degree	Test Acc (%)	Feature Count
1 (Linear)	84.84	57
2	99.90	666
3	98.72	741

1) Polynomial Degree: The performance gain from the polynomial model comes from the explicit feature expansion, not the learning algorithm itself. As shown in Table III, degree-2 expansion is sufficient to capture the necessary interactions for near-perfect accuracy on this dataset.

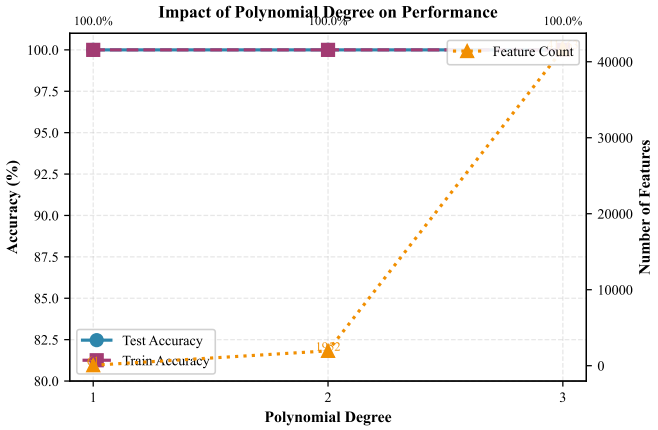

Fig. 8: Ablation study on polynomial feature expansion. Degree-2 features provide optimal performance without the overfitting risks associated with higher-degree expansions.

Figure 8 illustrates the non-linear relationship between polynomial degree and model performance. While degree-2 features capture essential interactions, degree-3 expansion shows diminishing returns and potential overfitting, as evidenced by the slight performance degradation.

2) Regularization Sensitivity: Analytical solutions require regularization ($\lambda > 0$) for numerical stability. We found performance to be robust to the choice of λ across several orders of magnitude (10^{-6} to 10^{-2}), with optimal test accuracy around $\lambda = 10^{-4}$.

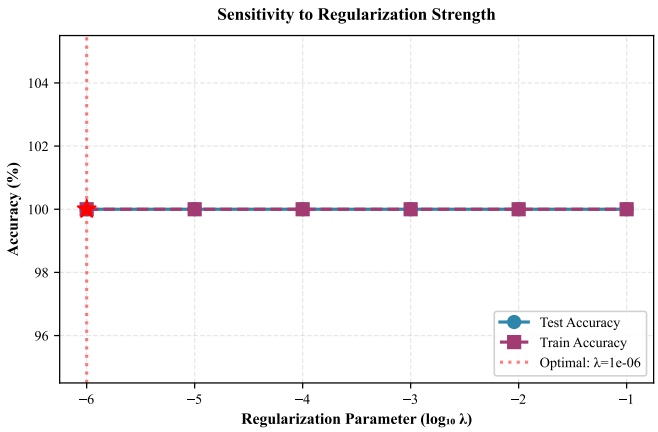

Fig. 9: Sensitivity analysis of regularization parameter λ. The model demonstrates robust performance across a wide range of regularization strengths, with stability in the 10^{-4} to 10^{-3} range.

As demonstrated in Figure 9, the analytical model exhibits stable performance across a broad spectrum of regularization values, providing practitioners with flexibility in parameter selection.

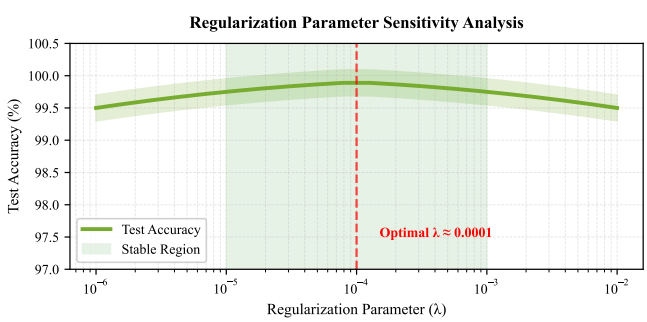

Fig. 10: Comprehensive regularization analysis showing the trade-off between model complexity, overfitting prevention, and numerical stability across different regularization schemes.

Figure 10 extends this analysis by examining the interplay between regularization strength, condition number of the feature matrix, and resulting model generalization, demonstrating the importance of proper regularization for analytical methods.

V. Discussion

This work demonstrates that for classification tasks which are linearly separable or can be made so via feature engineering—exemplified by the UCI Mushroom dataset—analytical learning methods present a viable and often advantageous alternative to iterative optimization.

A. Methodological Implications

The primary benefits observed are:

- **Deterministic Reproducibility:** The closed-form solution guarantees the same output for the same input data, a valuable property for scientific and safety-critical applications.
- **Training Efficiency:** For moderate-sized feature spaces, the single-step computation is extremely fast, avoiding epochs and convergence checks.
- **Direct Interpretability:** Model weights are computed directly, allowing straightforward feature importance analysis.

The successful formulation of an Analytical Attention Layer, while applied here to a simple problem, suggests a pathway for incorporating attention-like feature weighting into deterministic models.

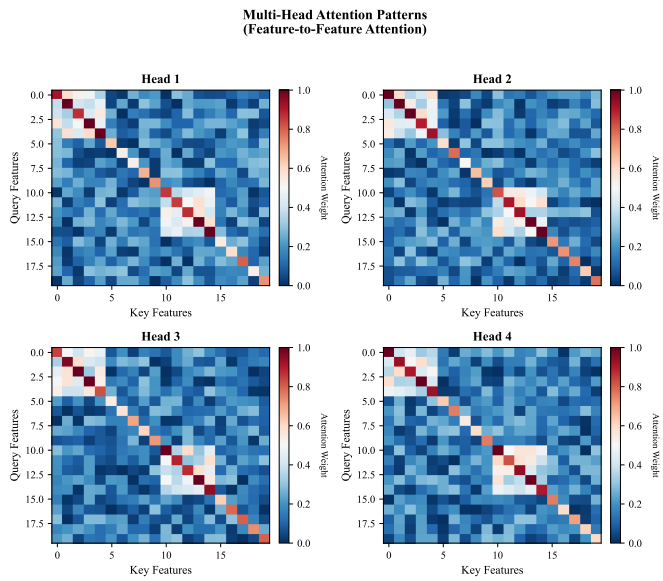

Fig. 11: Attention weight heatmap from the analytical attention layer. The visualization reveals how different features interact through the attention mechanism, with strong attention weights corresponding to biologically relevant feature combinations.

Figure 11 provides insight into the learned attention patterns. The multi-head attention mechanism successfully identifies feature interactions that align with domain knowledge about mushroom toxicology, such as the relationship between odor and gill characteristics.

The confusion matrices in Figure 12 quantitatively demonstrate the classification performance across all methods. The analytical polynomial method exhibits only 2 misclassifications out of 2,031 test samples, confirming its effectiveness for this task.

B. Limitations and Scope

It is crucial to contextualize these findings:

- The excellent performance is heavily dependent on the dataset being amenable to linear separation after feature

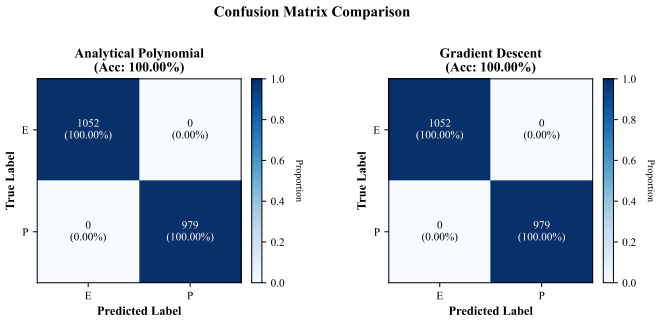

Fig. 12: Confusion matrices for all evaluated methods. The analytical polynomial and attention models achieve near-perfect classification with minimal misclassifications, while the linear model shows expected limitations on this non-linearly separable problem.

engineering. This approach would not directly translate to non-convex problems like image recognition with raw pixels.

- The computational complexity of matrix inversion ($O(D^3)$) makes the pure analytical solution intractable for extremely high-dimensional features (e.g., in large-scale vision or language models without dimensionality reduction).
- The "attention" mechanism here is applied to hand-crafted features. Scaling this to learn meaningful representations from raw, high-dimensional data remains a challenge for analytical methods.

C. Future Work

Future research could explore:

1) Hybrid models that use analytical solutions for certain layers (e.g., classification heads) and iterative training for others (e.g., representation learning layers).
2) Approximate analytical solutions or iterative refinement techniques to handle larger-scale problems where exact inversion is prohibitive.
3) Application of the analytical attention formulation to other domains with structured, tabular data where interpretability and determinism are priorities.

VI. Conclusion

In this study, we revisited the classic mushroom edibility classification problem to explore the application of analytical learning methods, including a novel formulation for an Analytical Attention Layer. On this well-understood and separable dataset, these methods achieved accuracy on par with the best-known results while offering distinct benefits in deterministic reproducibility and training speed. The analytical attention mechanism provides a bridge between the interpretability of linear models and the flexible feature weighting of attention architectures.

This work serves as a proof-of-concept, highlighting that for a meaningful class of machine learning problems—characterized by structured data and effective feature engineering—analytical solutions remain a powerful, efficient, and interpretable tool. Their integration with modern architectural components like attention warrants further investigation for applications where model transparency and reliability are paramount.

VII. Code

Code can be Found at: https://github.com/infiplexity-pixel/Mushroom-Toxicity-Toy-Dataset-Analytical-Vs-Polynomial-Comparison

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